Question Bank for Basics of civil and mechanical engineering subject's syllabus

Questions	Expected Marks
1. Which types of core concepts are studied in mechanical Engineering?	2
2. What is the role of mechanical engineer in engineering society?	2
3. What are the branches of mechanical engineering?	2,4
4. Define the thermodynamics.	2
5. Define System.	2
6. What is meant by surrounding in thermodynamics?	2
7. State the types of system boundary?	2
8. What are the types of system?	2
9. Define Enthalpy.	2
10. Define Entropy.	2
11. What is zeroth law of thermodynamics?	2
12. What is first law of thermodynamics?	2
13. What is second law of thermodynamics?	2
14. What is third law of thermodynamics?	2
15. Describe the laws of thermodynamics.	7

16. Define the heat engine & state its types?	4
17. What are different types of internal combustion engine?	2
18. Describe the main component of internal combustion engine with its neat labelled diagram.	7
19. What the function of flywheel in reciprocating IC engine.	2
20. What is the function of two stroke engine?	2
21. Differentiate between two stroke engine & four strokes engine.	4,7
22. What are the conventional energy source?	2
23. What are the Non-conventional energy source?	2
24. What is meant by conventional & non-conventional energy?	2
25. How steam turbine power works and draw its neat labelled diagram?	7
26. How gas turbine power plant works and draw its neat labelled diagram?	7
27. List the main components of gas turbine power station.	4
28. Describe the five segments of auto components with its neat labelled diagram?	7
29. What is meant by nuclear fusion?	4
30. Describe nuclear fission reaction.	4
31. What is law of conservation of energy?	2
32. How vehicle are classified on the basis of load?	2
33. Define automobile engineering.	2
34. How vehicle are classified on the basis of load?	2
35. How vehicle are classified on the basis of Wheels?	2
36. How vehicle are classified on the basis of Fuel used?	2
37. How vehicle are classified on the basis of vehicle body?	2

38.How vehicle are classified on the basis of Transmission?	2
39.How vehicle are classified on the basis of position of engine?	2
40.Define quality.	2
41.What is conformance to requirement?	2
42.What is meant by customer's preference?	2
43.What are two aspects of Quality management?	2
44.Define quality assurance.	2
45.Define quality control.	2
46.Explain in brief the quality control process.	4
47.Describe the production system with flow chart.	7
48.List the steps required in production system.	4
49.Describe the manufacturing by its processes.	4
50.Define machining.	2
51.Enlist the operations involves in machining processes.	2
52.What is the principal of machining with neat labelled diagram?	4
53.What are the essential requirements for machining?	4
54.Define machinability.	2
55. Describe the attempts are made to measure machinability.	4
56. Explain essential components of machining with neat & labelled diagram.	4
57.What are the machining parameter?	4
58.Define cutting speed.	2
59.Define feed.	2
60.Define depth of cut.	2

61. What are the physical functions of machine tool?	4
62. Explain lathe machine's main parts with its neat labelled diagram.	7
63. Define lathe machine.	2
64. Draw the neat labelled diagram of lathe machine.	4
65. What is the function of tailstock?	2
66. Where is the carriage part located in lathe machine?	2
67. Enlist the main parts of carriage.	2
68. Explain in details the main parts of carriage of the lathe machine.	4
69. What is the function of lead screw in lathe machine?	2
70. What are the different types of lathe machine?	2
71. How capstan and turret lathe works?	2
72. What is the function of drilling machine?	2
73. Define drilling operation.	2
74. How drilling machine works explain in details?	4
75. What are the different types of drilling machines?	2
76. Explain the main parts of sensitive drilling machine with its neat labelled diagram.	7
77. How the radial drilling machine works?	4
78. Describe the different operations that can be performed in a drilling machine.	7
79. Enlist the different drilling operation.	2
80. What is meant by Boring explain with neat labelled diagram?	4
81. What is meant by Reaming explain with neat labelled diagram?	4
82. What is meant by Tapping explain with neat labelled diagram?	4

83. Describe the milling machine with its neat labelled diagram.	7
84. What is the function of milling machine?	2
85. What are the different types of milling machines?	2
86. Define casting.	2
87. Explain sand casting process.	7
88. How the machine design is classified?	2, 4
89. What are the general considerations in machine design?	4
90. How engineering material is classified?	2
91. How the metals are classified?	2
92. Define factor of safety.	2
93. Explain any four (or seven) operations performed on lathe machine.	4, 7
94. Enlist the operations performed on lathe machine.	2
95. Which types of work is performed in civil engineering profession?	2, 4
96. State the various branches of civil engineering.	4
97. List any four civil engineering materials?	2
98. State the types of stones?	4, 7
99. How the geological rocks can be classified?	2
100. Describe in details the properties of stones required in civil construction work.	12
101. How the bricks are used in civil work?	2
102. How cement is made in cement making factory?	2
103. State the different types of cement?	7,12
104. Describe the physical properties of cement which are specified in IS 269-1967.	4

105. How the cement is use in various construction activities?	4
106. How sand obtained artificially?	2
107. What are the properties of good sand?	4
108. Enlist the elements of a building structure?	2
109. Describe the basic elements of a building structure.	7
110. List any four building components and planning material?	2
111. Discuss in details basic requirements for the planning and construction of a building structure?	12
112. How a building structure be planned for energy efficiency?	4
113. Define Beam in building structure.	2
114. How are classified and identify column of a building structure?	4
115. What are the various types of roofs?	2
116. State any eight differences between Load bearing structure and framed structure.	12
117. What is the importance of bridge structure?	2
118. State the various types of bridges?	7
119. How Bridge can be classified span length factor?	4
120. How bridge can be classified by relative position of bridge floor?	2
121. What are the main parts included sub structure of a bridge?	4
122. State the components of superstructure of bridge?	4
123. What is purpose of dam structure?	4
124. Classify the dam structure according to its use?	4
125. What is meant by surveying in civil engineering work?	2
126. Describe in details the purpose of survey in civil work?	7

127. What are the two fundamental principles of surveying?	4
128. Define levelling in construction work?	2
129. Which types of instruments are used for levelling?	2
130. What is the dumpy level?	4
131. What is meant by plane of collimation?	2
132. What is principal of levelling?	4